

Modeling the Impact of Basel III Capital Requirements on Bank Lending Rates

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Abstract

This study investigates the impact of the new capital requirements under the Basel III framework on bank lending rates. By constructing a stylized representative bank's financial statements, I show that the higher cost associated with a one-percentage increase in the capital ratio can be recovered by increasing the bank lending rate. The results indicate that in the case of scheduled commercial banks, one-percentage point increase in capital ratio can be recovered by a commensurate increase in the bank lending rate by 16 basis points and would go up to an extent of 94 basis points for six-percentage point increase assuming that the risk-weighted assets are unchanged. The results assume significance as the estimations for the scenarios of changes in risk-weighted assets change in return on equity (ROE) and the cost of debt.

JEL Classification: G21; G28; E44; E51; E61

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1. Introduction

The lessons from the analysis of global financial crisis underscore the role of bank regulation as vital because of its severe potential impact on economies. In the context of implementation of Basel III proposals proposed by [BCBS \(2010\)](#), on the one hand, there are logical points of view that higher capital requirements lead to (i) a healthier financial system through a reduced risk of bank failure, (ii) lower systemic risk, and (iii) lower social costs as a result of the elimination of moral hazard. On the other hand, the counter-arguments highlight that higher capital requirements will escalate the costs of financing for banks as equity financing is more expensive than debt financing, leading to (i) a slowdown in credit growth and the potential detrimental effects on the economies concerned, and (ii) a decrease in profitability, preventing future accumulation of capital via retained earnings and ruling out future expansion of banks' operations ([Cosimano and Hakura, 2011](#)). However, the overall objective of Basel III is to strengthen the global capital, liquidity, and risk assessment rules and enhance the resilience of the banking sector.

The higher capital requirements cause an increase in the weighted average cost of capital by raising the marginal cost of funding and consequently banks would have to pass on the increase in the cost of capital to the borrowers as higher lending rates¹. Per se, the cost of lending might get higher in case that the cost of equity financing becomes higher than the cost of debt financing; such circumstance shall reduce the rate of bank financing and of course it would slow down any forecast for economic growth ([Angelini et al., 2011](#)). However, the important question is how much would be the increase in the bank lending rates and how can these interrelations be mapped appropriately for estimation? Bank for International Settlements (BIS) estimates in general for a representative bank observe that one-percentage point increase in the capital ratio raises loan rates by 13 basis points ([BCBS, 2010](#)). Besides, the additional cost of meeting the liquidity standard amounts to around 25 basis points in lending rates when risk-weighted assets are unchanged. On the other hand, the lending rates drop to 14 basis points or less after considering the fall in risk-weighted assets and the corresponding lower regulatory capital needs associated with the higher holdings of low-risk assets.

¹ Higher capital requirements will increase the banks' marginal cost of loans if the marginal cost of capital is greater than the marginal cost of deposits", i.e. if there is a net cost of raising capital.

Though literature is rich with studies on bank regulation and its impact, very few studies (BIS 2010, King 2010; Slovik and Courneade 2011; Angelini et al., 2011) are reported on the modeling of the impact of Basel III induced higher capital requirements on bank lending rates. Country-specific studies are required to estimate the impact of new Basel proposals and provide the much-needed logic and rational approach towards ascertaining these impacts to policy makers as well as build a body of literature capturing the uniqueness of country specific features. Regardless of having low or almost nil exposure to the toxic assets involved in the global financial crisis and a gradualist approach towards liberalization of the financial sector, the Indian financial sector was indeed affected by the global financial crisis. The Basel III norms are bound to have a significant impact on the Indian financial sector. Although it is felt to be in a comfortable position to meet some of the Basel III norms, the implementation of some of the other norms will be a challenge. With this motivation, this study estimates the impact of Basel III proposals, particularly with regard to the impact on bank lending rates as this would have a direct effect on the economic output.

The design of the representative bank model is useful in mapping the changes in the bank's capital structure and understanding as to how the composition of assets has an effect on the different components of net income using the standard accounting relationships. Even though banks can adjust to the regulatory reforms in several ways, this study supposes that they seek to pass on any additional cost by raising the cost of loans to borrowers. I assume that by computing the change in net income and shareholder's equity associated with the regulatory changes, I can compute the increase in lending rates required to achieve a given return on equity (ROE). In view of the enormous significance of the impact of Basel III on banks, this research outcome benefits the practitioners in the industry and researchers apart from contributing to the literature on bank regulation and risk management with a newer and topical approach for quantification of the impacts of new regulatory standards.

This approach is illustrative and general in nature and could be used to estimate the impact on lending rates from a change in bank's capital structure, asset composition, risk-weighted assets and the corporate tax paid by these banks. This approach is acceptable, particularly for practitioners for easy comprehension. The added benefit of this approach is that it explains how a given change can alter the bank's profitability and indicates different

possible behavioral responses to the regulations including the unintended consequences. Further, this approach being a bottom-up, micro-founded one, it offers a useful complement to top-down, structural models where the modeling of the financial sector is necessarily parsimonious. Another contribution of this study is that it considers three different groupings of banks: (i) scheduled commercial banks; (ii) public sector banks; and (iii) private banks in Indian banking.

The remainder of the paper is organized as follows. Section 2 presents the related literature and Section 3 describes the model based on the theoretical considerations. Section 4 provides the description of the data employed for the analysis. Section 5 presents the results and discussion. Section 6 concludes.

2. Related Literature

Though there is a wealth of literature on bank capital and regulation (refer [Dewatripont and Tirole, 1994](#); [Bhattacharya *et al.*, 1998](#); [Santos, 2001](#); and [VanHoose, 2007](#)), very few studies have highlighted on the aspect of loan pricing. A strand of literature that handled the effects of capital requirements upon banking performance suggest that higher capital requirements affect the profitability of commercial banks ([Ferri, and Majnoni, 2001](#); [Rojas-Suarez, 2002](#)). Another strand of literature argues in favor of higher capital requirements as they are found to strengthen the banks and minimize the risk of bankruptcy considerably ([Admati *et al.*, 2010](#))

[Repullo and Suarez \(2004\)](#) analyze the loan pricing implications of capital requirements in a credit market where loan default rates are driven by a single factor of systematic risk. Their loan pricing equation entails that low-risk firms will accomplish decreases in their loan rates by borrowing from banks adopting the IRB approach, while high-risk firms will evade increases in their loan rates by borrowing from banks that embrace the less risk-sensitive standardized approach of Basel II. However, they contemplate a perfectly competitive market for business loans, where a single risk factor expounds the correlation in defaults across firms.

Another recent study in this direction is by [Ruthenberg and Landskroner \(2008\)](#), who estimate the possible effects of the new rules on the pricing of bank loans using Israeli

economic data. Their loan equation is founded on a model of a banking firm facing uncertainty operating in an imperfectly competitive loan market. They show that high-quality corporates and retail customers will experience a decline in loan interest rates in (large) banks that, will embrace the IRB approach. On the other hand, high-risk customers will benefit by shifting to (small) banks that will assume the Standardized approach. This study is distinctive in their approach by estimating the loan rates with loan rate equation involving the explanatory variables such as; loan loss provisions, market share, the cost of debt (secondary market), and cost of equity sensitivity.

Some studies report the impact of higher capital requirements on lending rates and volumes of lending. [Kashyap, Stein, and Hanson \(2010\)](#) study the effect of equity-to-asset ratio under the base III regulations and observe that the lending rates shall be up 6 basis points (bps). Similarly, [BIS \(2010\)](#) reports an even higher increase that shall take effect (in the range of 12.2 to 15.5 bps) after collecting data from 38 models, ran from national central banks of advanced economies. [Angelini et al., \(2011\)](#) report similar results. [Slovik and Cournede \(2011\)](#) employ accounting relations to report the results in an upper bound of 15 bps.

By providing an accounting-based analysis, [Elliott \(2010\)](#) estimates how much the interest rate charged on loans would increase if banks are required to hold more equity. However, in the stylized model of Elliot, banks hold only loans funded by equity, deposits and wholesale funding and the interest rate on loan is priced in order to meet a targeted ROE after covering the cost of liabilities and other fixed expenses (such as administrative costs and expected loan losses). Using the Federal Deposit Insurance Corporation (FDIC) data for aggregate United States (US) banking system, Elliott reports that if the ratio of common equity required for a given loan is raised by 2% with no other changes, banks would need to raise lending rates by 39 basis points (bps) to maintain the target ROE of 15%. Further, if the ROE is allowed to fall to 14.5%, the lending rate would have to rise only by 9 bps. Elliott summarizes that through a combination of actions the US banking system would be able to adjust to higher capital requirements and ensure that they would not have a strong effect on the pricing or availability of bank loans. The merit of Elliott's method is in its simplicity as well as the intuition, it provides on the pricing of loans and the alternatives available to banks to adjust to higher capital levels. Drawing from the literature discussed, I develop and present our model in the ensuing section.

3. The Model

The purpose of this section is to map the capital and liquidity requirements as per Basel III² to bank lending rates. This estimation supposes that the return on equity (ROE) and the cost of debt are unaffected, with no change in other sources of income and on the same line of thought; it is further assumed that there is no reduction in operating expenses. Such a mapping endows researcher with a useful instrument to analyze the impact of regulatory changes on the cost of credit and the real economy. A rise in the interest rate charged on bank loans is believed to reduce loan demand, all else equal, leading to a drop in investment and output. This methodology has been employed in the BCBS's assessment of the long-term economic impact of the proposed regulatory changes on output (BCBS, 2010, King, 2010). Further, the benefit of these estimates of changes in bank lending rates could be found in using them as inputs into dynamic stochastic general equilibrium models that have been augmented to include a micro-founded banking sector such as Goodfriend and McCallum (2007), or as a proxy for increased financial frictions in macroeconomic models that lack a financial sector. Motivated by similar studies such as Repullo and Suarez (2004) and Ruthenberg and Landskroner (2008) for the Basel II framework, this mapping exercise illustrates the potential implications of Basel III higher capital requirements on bank lending rates.

A *representative bank*³ is designed to map the changes in the bank's capital structure and to understand how the composition of assets has an effect on the different components of net income using the standard accounting relationships. Even though banks can adjust to the regulatory reforms in several ways, this study supposes that they seek to pass on any additional cost by raising the cost of loans to end-customers. It is believed that by computing the change in net income and shareholder's equity associated with the regulatory changes, I can compute the increase in lending rates required to achieve a given return on equity (ROE).

This approach being a bottom-up, micro-founded one, it offers a useful complement to top-down, structural models where the modeling of the financial sector is necessarily parsimonious. Although this approach is founded on several assumptions, all the assumptions are apparent, realistic, and simple and can be modified to check the sensitivity of the results.

² The 1988 Basel Accord is referred to, as Basel I and the 2004 revision of the same is known as Basel II.

³ The author is grateful to Michel R King for suggesting the suitability of this approach

This approach focuses on only two elements of Basel III proposals: (i) relating to raising the minimum capital requirement and (ii) relating to enhanced liquidity requirement. Firstly, though the previous Basel accords (Basel I and II) specified capital adequacy rules for minimum capital adequacy ratios, however, they could not absorb the losses during the recent crisis. In this backdrop, Basel III stipulates higher levels of tangible common equity. In order to achieve this, banks need to increase their common equity, with high-quality capital.

This approach, of course, is not without limitations as it does not formally model the choices faced by the banks, nor does it offers estimates based on an optimization in a general equilibrium setting. On the other hand, as a substitute, it offers a starting point for understanding the behavioral response of banks to a regulatory change in a most acceptable practical setting. It enables the researchers and the policy makers in determining the impact given a country's institutional setting, its banking sector and the elasticity of loan demand. Though this approach can suggest the potential magnitude of the change in lending rates, deciding whether banks would be able to pass on these costs to borrowers is beyond the scope of this study. Further, this approach focuses on the 'steady state' and does not consider the transition period to the higher regulatory requirements⁴. In the steady state, the supply of bank credit is considered as exogenous and credit rationing is ignored. It is further implied that banks price their loans to meet the marginal cost of loan production.

This study does not focus on measurement of credit risk, but on the relationships between a bank's capital structure, asset composition and their impact on bank's profitability. This greater level of detail is vital for understanding as to how the banks respond to the Basel III regulatory reforms. Both theorists and researchers are quite concerned with understanding these relationships, albeit they may be too complex to model parsimoniously. The approach of this study is influenced largely by [Elliott \(2010\)](#). By the actual usage of the balance sheet data to compute the regulatory impact, it takes into account the composition of the assets and liabilities as well the very important distinction between the assets and risk-weighted assets. This study compares two steady states, namely, one with and another without the regulatory requirements.

⁴ The transition to the higher capital and liquidity requirements is the focus on the interim report of the Macroeconomic Assessment Group, [BIS \(2010\)](#).

I estimate the impact of Basel III higher capital requirements on bank lending rates is estimated by employing the OECD approach, wherein it is based on accounting identities applied to stylized banking sector balance sheets. Two categories of bank assets are considered for the purpose of the assessment: (i) lending assets of the banks (LA) that comprise bank's loans and advances to households and non-financial corporations held on banking books, and (ii) Investments and Other assets of the banks (IOA), which represent a residual category that comprises assets held on trading books, inter-bank assets, government bonds and other remaining assets. Further, one of the significant assumptions is that a bank can directly affect the pricing of LA by adjusting its lending rate. The pricing of IOA is mainly market driven and it is therefore assumed that a bank cannot directly affect the pricing of these assets.

The balance-sheet identity presented in Eq (1) postulates that the return on bank assets is equal to bank funding costs, which are determined by the cost of liabilities and the cost of equity.

$$r^{LA}_t \times LA_{it} + r^{IOA}_t \times IOA_{it} = r^{Liab}_t \times Liab_{it} + r^E_t \times E_{it} \quad \text{----- Eq (1)}$$

Equation (2) incorporates a one-percentage point increase in bank capital relative to risk-weighted assets. An increase in bank capital will affect bank liability and equity structures and as a consequence the overall bank funding cost.

$$r^{LA}_{t+1} \times LA_{it} + r^{IOA}_t \times IOA_{it} = r^{Liab}_t \times (Liab_{it} - \frac{RWA_{it}}{100}) + (r^E_t \times \frac{RWA_{it}}{100}) \quad \text{----- Eq (2)}$$

To maintain equilibrium for the change in funding cost, banks are assumed to adjust their lending rates, while their costs of equity and debt financing are assumed to remain constant (Eq 3).

$$(r^{LA}_{t+1} - r^{LA}_t) = (r^E_t - r^{Liab}_t) / LA_{it} \times \frac{RWA_{it}}{100} \quad \text{----- Eq (3)}$$

Combining equations (1) and (2) leads to equation (3), which shows the increase in bank lending rates as a result of a one-percentage point increase in the ratio of bank capital to risk-weighted assets: where;

LA_{it}	is the <i>loans and advances to total assets</i>
IOA_{it}	is the <i>investments and other assets to total assets</i>
$Liab_{it}$	is the <i>liabilities to total assets</i>
E_{it}	represents <i>common equity to total assets</i>
RWA_{it}	represents <i>risk-weighted assets to total assets</i>
r^{LA}_t	represents <i>return on advances</i>
r^{IOA}_t	represents <i>return on investments and other assets</i>
r^{Liab}_t	represents <i>cost of borrowing</i>
r^E_t	represents <i>cost of equity</i>

All the variables are measured in percentages.

4. Data

For constructing a representative bank's financial statements, the stylized balance sheet and income statement data for scheduled commercial banks (SCBs) in India is collected from the Capitaline Plus data base⁵, respective websites of the banks under study, RBI database, and Basel II Pillar III disclosures of respective banks from their websites. The data employed for analysis are for the period from 2002 to 2011 as the Capitaline Plus database provides complete datasets only for a ten-year period. IDBI Bank in spite of its dual nature, it is also included amongst the public sector banks. The public sector banks (PSBs) category also includes the State Bank of India (SBI) and its associate banks. The private banks also include the foreign banks operating in India. The SCBs include both the public and private banks.

[Table 1 is about here]

This study employs the consolidated entity wherever available, taking the last filing in a calendar year. [Table 1](#) provides the details of banks by category and year. The number of banks varies by year due to the merger, closure, or entry of a new foreign bank in the year. The Capitaline Plus database does not report RWAs directly in its data sets. Instead, the quantity of RWAs is collected from the more authenticate source; i.e the Basel II Pillar III disclosures of respective banks from their websites. Since the Capitaline Plus database

⁵ Capitaline Plus provides fundamental and market data on more than 20,000 Indian listed and unlisted companies, classified under more than 300 industries, along with powerful analytic tools. www.capitaline.com

presents the data for the category of banks, there was no problem of the outliers and the requirement of winsorization of capital adequacy ratios to reduce the impact of outliers. Based on the data described, a representative bank balance sheet and income statement is constructed for each category of banks by taking the weighted average values for each component of the balance sheet and the income statement for banks in each category of study. The weights are based on total assets of the category of the banks⁶. Table 2 shows the stylized balance sheet and income statement for the representative bank of SCBs and all components are shown as a percentage of total assets.

[Table 2 is about here]

The loans & advances represent about more than half of the typical bank's assets, followed by investments & securities (31.48%), cash and balances with RBI (5.59%), other assets (4.63%) and inter-bank claims (4.09%). These assets are funded mainly by deposits (75.65%), bank borrowings or inter-bank funding (9.6%). The shareholder's equity is, to the extent of 7.12%. The RWAs constitute around 65.77% total assets on average. This ratio is significant when calculating the cost of meeting the higher capital requirement. Table 2 also shows the consolidated income statement for the representative bank. In terms of the composition of net income, net interest income is 2.24% with non-interest income also important at 1.28%. The total operating expenses constitute 2.32% of total assets. The personnel expenses represent around 41% of total operating expenses. The return on assets is 0.8%, implying that the average return on equity is around 15%. The average historical corporate tax rate is accepted at 33%.

5. Results and Discussion

The experiment assumes that the capital ratio is raised by increasing equity and reducing long-term debt correspondingly. Especially, it is assumed that (i) any higher cost of funding associated with this change is *fully recovered* by raising loan rates – 100% pass through; and (ii) that the costs of equity and of debt are *not* affected by the lower riskiness of the bank. As discussed in more detail below, this, together with the rather conservative assumption about the initial ROE, suggests that the results should be viewed as providing something close to an upper bound of the impact on loan rates. The results are obviously sensitive to a host of the assumptions in the analysis. Banks could offset the loss of net income arising from meeting increased capital requirements through other means than raising

⁶ All variables are standardized by dividing by bank's total assets in each year

loan rates. For example, banks could in principle (i) increase non-interest income (e.g. fees and commissions), (ii) reduce the rate paid on deposits, or (iii) reduce operating expenses. Any combination of these actions will generate higher net income and reduce the need to raise lending rates. We can also provide a sense of the magnitudes involved. The rise in lending rates associated with a 1-percentage point increase in the capital ratio could be avoided by reducing operating expenses too.

Undoubtedly, there are enough reasons to believe that the cost of capital would *decline* in response to a reduction in bank leverage. As capital levels rise and the bank turns out to be safer, both of these costs tend to decline by further decreasing the impact on lending rates. Moreover, the modification in the cost of capital may perhaps reduce to tax effects (Modigliani and Miller, 1958). However, such a scenario has not been studied in arriving at the estimates. After having outlined the methodology and data employed in the computation of the stylized bank balance sheet and the income statement, this section provides the results of the estimations of the potential impact of raising the capital ratio on lending rates.

[Table 3 is about here]

The estimated sensitivities of bank lending rates to a one-percentage-point increase in capital requirements in Table 3 suggest that in the case of SCBs, 1-percentage point increase in the ratio of capital to risk-weighted assets will push up bank lending rate by 15.63 basis points (bps) given that RWAs are unchanged. However, the bank lending rate rises by 21.22 bps when the RWAs are assumed to increase by 20 percentage points. Similarly, for a 2-percentage point increase in capital ratio, bank lending rate would rise by 31.26 bps when RWAs are unchanged and would increase by 42.44 bps when RWAs are assumed to increase by 20 percentage points. For a 3-percentage point increase in CAR, the bank-lending rate would rise by 46.89 bps given that RWAs are unchanged and 63.66 bps when RWAs are increased by 20 percentage points. On the same lines, it was observed that for an increase of 4, 5 and 6-percentage points in the ratio of capital to risk-weighted assets, the bank-lending rate would increase by 62 bps, 78 bps, and 93 bps given that RWAs are unchanged and would rise by 85 bps, 106 bps, and 127 bps when RWAs are assumed to decrease by 20 percentage points.

The estimated sensitivities (for scenarios; RWAs unchanged, increase in RWAs and decrease in RWAs) of bank lending rate to a one percentage point increase in capital

requirements for scheduled commercial banks are presented in Table 4. The rise in bank lending rate in the case of SCBs has to be 15.63 bps, 21.22 bps, and 15.01 bps in the scenarios; RWAs unchanged, with a 20% increase in RWAs, and with a 20% decrease in RWAs respectively.

[Table 4 is about here]

For PSBs, a one-percentage point increase in the ratio of capital to risk-weighted assets will increase the lending rate by 12.48 bps on average given that RWAs are unchanged. However, the results reported in Table 5 show that the bank lending rate declines by 10 bps when RWAs are assumed to decline by 20% and rise to 14.98 bps when RWAs are assumed to increase by 20%.

[Table 5 is about here]

The estimated sensitivities for scenarios like; RWAs unchanged, increase in RWAs and decrease in RWAs (Table 6) indicate the bank lending rate to a one percentage point increase in capital requirements for public sector banks. The rise in bank lending rate in the case of SCBs has to be 12.48 bps, 14.98 bps, and 9.99 bps in the scenarios; RWAs unchanged, with a 20% increase in RWAs, and with a 20% decrease in RWAs respectively.

[Table 6 is about here]

For private banks, a one-percentage point increase in the ratio of capital to risk-weighted assets will raise the lending rate by 22.48 bps on average given that RWAs are unchanged (Table 7). However, the bank lending rate declines to 17.98 bps when RWAs are assumed to decline by 20% and rise to 26.98 bps when RWAs are assumed to increase by 20%.

[Table 7 is about here]

The estimated sensitivities (for scenarios; RWAs unchanged, increase in RWAs and decrease in RWAs) of bank lending rate to a one percentage point increase in capital requirements for scheduled commercial banks are presented in Table 8. The rise in bank lending rate in the case of private banks has to be 22.48 bps, 26.98 bps, and 17.98 bps in the scenarios; RWAs unchanged, with a 20% increase in RWAs, and with a 20% decrease in RWAs respectively.

[Table 8 is about here]

In order to present a comparison of increase in bank lending rates (when RWAs are unchanged) for a one percentage point increase in bank capital of scheduled commercial

banks, public sector banks and private banks, Table 9 illustrates that private banks experience a significant rise in lending rate compared to other bank categories.

[Table 9 is about here]

Similarly, on a comparison of increase in bank lending rates (when RWAs are increased by 20%) for a one percentage point increase in bank capital of scheduled commercial banks, public sector banks and private banks, Table 10 shows that private banks experience higher lending rate when compared other categories of banks.

[Table 10 is about here]

In the same way, on a comparison of increase in bank lending rates (when RWAs are decreased by 20%) for a one percentage point increase in bank capital of scheduled commercial banks, public sector banks and private banks, Table 11 shows that private banks continue to experience higher lending rate compared other categories of banks.

[Table 11 is about here]

The sensitivity is found to be comparatively greater in the United States (mainly due to a higher return on equity and a higher share of risk-weighted assets on bank balance sheets) and lower in Japan (mostly due to a lower return on equity and a higher share of lending assets on bank balance sheets) (refer Table 12). The increase in bank lending rate for a one percentage point increase in bank capital in the Euro area is 14.3 percent and it is 15.63 percent in India.

[Table 12 is about here]

6. Conclusion

The paper provides an evidence for the modeling the impact of Basel III capital requirements on bank lending rates using the Indian banking data. Employing a micro-founded bottom-up approach, it offers a useful complement to top-down, structural models where the modeling of the financial sector is necessarily parsimonious. It considers three different groupings of banks: (i) scheduled commercial banks; (ii) public sector banks; and (iii) private banks in Indian banking and provides the estimates of the rise in lending rates due to the impact of higher capital requirements. For one-percentage point increase in the capital ratio, the lending rates of SCBs, PSBs, and private banks, rise by 15.63 bps, 12.48 bps, and 22.48 bps respectively when RWAs are unchanged. Assuming 20 percent decline in RWAs, the lending rates of SCBs, PSBs, and private banks, rise by 15.01 bps, 9.99 bps, and 17.98 bps respectively. However, in the scenario, assuming a 20 percent increase in RWAs,

the lending rates of SCBs, PSBs, and private banks, rise by 21.22 bps, 14.98 bps, and 26.98 bps respectively. The variance in the impact reflects the differences in the capital structure and loan demand of the different groupings of banks.

On a comparison of increase in bank lending rate for a one-percentage point increase in the capital ratio of bank groups, private banks experience significant rise. Further, on a comparison of the estimates for USA, Euro area, Japan and India, the sensitivity is found to be comparatively greater in the United States (mainly due to a higher return on equity and a higher share of risk-weighted assets in bank balance sheets) and lower in Japan (mostly due to a lower return on equity and a higher share of lending assets on bank balance sheets). The variance in the results reflects cross-country differences in the elasticity of loan demand with respect to lending rate and banks' net cost of raising equity.

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Table 1: Sample distribution by category of SCBs in India

Year	Public Sector Banks	Private Banks	SCBs
2002	30	87	117
2003	30	94	124
2004	30	93	123
2005	30	97	127
2006	30	88	118
2007	30	86	116
2008	30	83	113
2009	29	69	98
2010	29	87	116
2011	28	85	113

Table 2: Stylized Balance Sheet and Income Statement of SCBs

Balance Sheet	Average	Income Statement	Average
Cash & balances with RBI	5.59	Interest income	6.21
Inter-bank claims	4.09	Interest expended	3.97
Investments & securities	31.48	A. Net interest income	2.24
Loans & advances	53.23	B. Other income	1.28
Fixed assets	0.99	C. Total revenues (A+B)	3.52
Other assets	4.63	D. Personnel expenses	0.95
TOTAL ASSETS	100.00	E. Other admin expenses	1.37
Deposits by customers (retail and corporate)	75.65	F. Operating expenses (D+E)	2.32
Inter-bank funding (borrowing)	9.16	G. Operating profit (C-F)	1.20
Other liabilities & provisions	8.06	H. Tax provision	0.36
TOTAL LIABILITIES	92.88	I. Net income (G-H)	0.84
Capital	1.05		
Reserves total	6.06		
Equity share warrants	0.00		
Equity application money	0.00	Return on equity (ROE) (%)	0.15
Total capital	7.12	Leverage multiple	6.60
TOTAL LIABILITIES & CAPITAL	100.00		
Risk weighted assets / Total assets	65.77	Average effective tax rate	33.00%

Table 3: Impact of increase in capital ratio on lending rates for SCBs

Increase in capital ratio (percentage point)	Change in lending rate to meet the cost of capital (bps)		
	Assuming RWAs unchanged	Assuming 20% decline in RWAs	Assuming 20% increase in RWAs
1	15.63	15.01	21.22
2	31.26	30.02	42.44
3	46.89	45.03	63.66
4	62.52	60.04	84.88
5	78.15	75.05	106.10
6	93.78	90.06	127.32

Table 4: Increase in bank lending rate for a one percentage point increase in bank capital for SCBs

Scenario	LA (percentages)	RWAs (percentages)	$(r^{LA}_{t+1} - r^{LA}_t)$ (basis points)
RWAs unchanged	53.22	65.77	15.63
with 20% increase in RWAs	53.22	78.92	21.22
with 20% decrease in RWAs	53.22	52.61	15.01

Table 5: Impact of increase in capital ratio on lending rates for PSBs

Increase in capital ratio (percentage point)	Change in lending rate to meet the cost of capital (bps)		
	Assuming RWAs unchanged	Assuming 20% decline in RWAs	Assuming 20% increase in RWAs
1	12.48	9.99	14.98
2	24.96	19.98	29.96
3	37.44	29.97	44.94
4	49.92	39.96	59.92
5	62.4	49.95	74.90
6	74.88	59.94	89.88

Table 6: Increase in bank lending rates for a one percentage point increase in bank capital for PSBs

Scenario	LA (percentages)	RWAs (percentages)	$(r^{LA}_{t+1} - r^{LA}_t)$ (basis points)
RWAs unchanged	54.73	52.54	12.48
with 20% increase in RWAs	54.73	63.04	14.98
with 20% decrease in RWAs	54.73	42.03	9.99

Table 7: Impact of increase in capital ratio on lending rates for Private Banks

Increase in capital ratio (percentage point)	Change in lending rate to meet the cost of capital (bps)		
	Assuming RWAs unchanged	Assuming 20% decline in RWAs	Assuming 20% increase in RWAs
1	22.48	17.98	26.98
2	44.96	35.96	53.96
3	67.44	53.94	80.94
4	89.92	71.92	107.92
5	112.40	89.9	134.90
6	134.88	107.88	161.88

Table 8: Increase in bank lending rates for a one percentage point increase in bank capital for Private Banks

Scenario	LA (percentages)	RWAs (percentages)	$(r^{LA}_{t+1} - r^{LA}_t)$ (basis points)
RWAs unchanged	49.53	96.09	22.48
with 20% increase in RWAs	49.53	115.38	26.98
with 20% decrease in RWAs	49.53	76.87	17.98

Table 9: Comparison of increase in bank lending rate for a one percentage point increase in bank capital (with RWAs unchanged)

	SCBs	PSBs	Private Banks
$(r^E_t - r^{Liab}_t)$ (basis points)	12.65	12.65	11.59
LA (percentages)	53.22	54.73	49.53
RWAs (percentages)	65.77	52.54	96.09
$(r^{LA}_{t+1} - r^{LA}_t)$ (basis points)	15.63	12.48	22.48

Table 10: Comparison of increase in bank lending rate for a one percentage point increase in bank capital (with 20% increase in RWAs)

	SCBs	PSBs	Private Banks
$(r_t^E - r_t^{Liab})$ (basis points)	12.65	12.65	11.59
LA (percentages)	53.22	54.73	49.53
RWAs (percentages)	78.92	63.04	115.38
$(r_{t+1}^{LA} - r_t^{LA})$ (basis points)	15.63	14.98	26.98

Table 11: Comparison of increase in bank lending rate for a one percentage point increase in bank capital (with 20% decrease in RWAs)

	SCBs	PSBs	Private Banks
$(r_t^E - r_t^{Liab})$ (basis points)	12.65	12.65	11.59
LA (percentages)	53.22	54.73	49.53
RWAs (percentages)	52.61	42.03	76.87
$(r_{t+1}^{LA} - r_t^{LA})$ (basis points)	15.01	9.99	17.98

Table 12: Increase in bank lending rate for a one percentage point increase in bank capital

	USA*	Euro Area*	Japan*	India ^v
$(r_t^E - r_t^{Liab})$ (basis points)	12.7	9.4	7.7	12.65
LA (percentages)	47.5	35.4	66.0	49.53
RWAs (percentages)	76.4	53.9	72.0	65.77
$(r_{t+1}^{LA} - r_t^{LA})$ (basis points)	20.5	14.3	8.4	15.63
bank lending rates				

Note: * OECD study (Slovik and Cournède, 2011)

^v this study